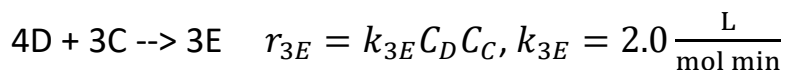
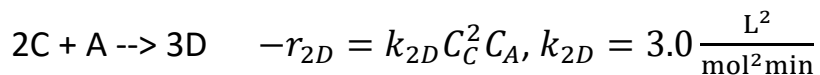
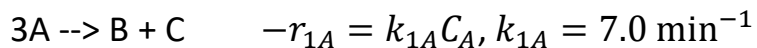


E08 Multiple Reactions in an Isothermal CSTR, 2 ways

The following reactions are taking place in liquid phase in a CSTR:



Write expressions for the net rates of A, B, C, D, and E, and use them to calculate the steady state rates in a CSTR that is operating at $C_A = 0.10 \text{ M}$, $C_B = 0.93 \text{ M}$, $C_C = 0.51 \text{ M}$, $C_D = 0.049 \text{ M}$.

1. Determine the CSTR volume.

We get the volume from

Here we have 5 species, we can use the mole balance for

At first glance, species E might be a good choice, since

However,

Looking again, species

Writing the net rate of species B and solving:

2. Let's pretend we don't know any of the exit concentrations, but we know the volume is 400 L. Determine the exit concentrations.

Now, the number of unknowns =

We hence need 5 equations. The equations that we will use are the

These take the form

Alternatively, we can write them

Setting them equal to zero sets us up well to use a

To do this, first let's write the

Decision Guide: Multiple Reactions in an Isothermal CSTR

Goal

For a multiple-reaction CSTR:

Part 1: Determine the **CSTR volume** when the exit concentrations are known.

Part 2: Determine the **exit concentrations** when the reactor volume is known.

Part 1: Determine the Reactor Volume

Step 1: Identify What Is Known

The problem provides:

- Feed concentration
- Volumetric flowrate
- Exit concentrations

and asks for: Reactor Volume

Step 2: Calculate All Reaction Rates

Use the provided rate laws. All reaction rates should be evaluated using the known reactor-exit concentrations.

Reaction 1



Rate law:

$$-r_{1A} = k_{1A} C_A$$

where

$$K_{1A} = 7.0 \text{ min}^{-1}.$$

Reaction 2



Rate law:

$$-r_{2A} = k_{2D} C_C^2 C_A$$

where

$$k_{2D} = 3.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}.$$

Reaction 3



Rate law:

$$r_{3E} = k_{3E} C_C C_D$$

where

$$k_{3E} = 2.0 \text{ L mol}^{-1} \text{ min}^{-1}.$$

Why Are We Calculating These First?

The CSTR mole balance requires a species rate. However, species rates are built from the individual reaction rates. Workflow:

Calculate Reaction Rates

↓

Calculate Net Species Rates

↓

Apply CSTR Mole Balance

↓

Solve for Reactor Volume

Concept Check: In multiple-reaction systems, always calculate the individual reaction rates first. Those reaction rates are then combined to obtain the net production or consumption rate of each species. Evaluate all reaction rates using the known exit concentrations.

Step 3: Choose a Species Balance

For a CSTR:

$$V = \frac{F_{i0} - F_i}{-r_i}$$

Use the species whose net rate is easiest to calculate.

Step 4: Calculate the Net Species Rate

In multiple-reaction systems: Species Rate $\neq$ Reaction Rate

A species may be:

- Produced in one reaction
- Consumed in another reaction

Therefore, calculate the **net rate of the species** before using the CSTR mole balance.

Step 5: Calculate the Reactor Volume

Substitute:

- Feed conditions
- Exit concentrations
- Net species rate

into the CSTR mole balance to obtain the reactor volume.

Part 2: Determine the Exit Concentrations

Step 1: Count Unknowns

If the reactor volume is known but concentrations are unknown:

Unknowns: CA, CB, CC, CD, CE

Total unknowns: 5

Step 2: Count Equations

The number of equations must equal the number of unknowns.

Therefore, we need: 5 independent equations

The most natural choice is: One species balance for each species. Because the unknowns are: CA, CB, CC, CD, CE

We write: Species A Balance, Species B Balance, Species C Balance, Species D Balance, Species E Balance

This gives: 5 equations, 5 unknowns

which means the system can be solved.

Why Is This Step Important?

Before attempting to solve any reactor-design problem, always check that: Number of Equations = Number of Unknowns. If there are fewer equations than unknowns, additional information or assumptions are needed. If there are more equations than unknowns, some equations may be redundant or inconsistent.

Concept Check: In Part 2, the unknowns are the five exit concentrations. Therefore, we need five independent equations. The most natural choice is one species balance for each species present in the reaction network.

Step 3: Write the Species Balances

At steady state:

using the general steady-state mole balance. Write one balance for: A, B, C, D, E

This gives: Species A Balance, Species B Balance, Species C Balance, Species D Balance, Species E Balance

Each balance relates the species inlet flowrate, outlet flowrate, and net rate of production or consumption within the reactor.

Why Do We Need Five Species Balances?

The unknowns are: CA, CB, CC, CD, CE

Since there are five unknown concentrations, we need five independent equations. The species balances provide those equations.

Step 4: Express Species Rates in Terms of Reaction Rates

Construct a species-accounting table.

Species A

Consumed in: Reaction 1, Reaction 2

Species B

Produced in: Reaction 1

Species C

Produced in: Reaction 1

Consumed in: Reaction 2, Reaction 3

Species D

Produced in: Reaction 2

Consumed in: Reaction 3

Species E

Produced in: Reaction 3

Use these relationships to write: r_A , r_B , r_C , r_D , r_E
in terms of the underlying reaction rates.

Why Is This Step Necessary?

The mole balances require species rates: r_A , r_B , r_C , r_D , r_E . But the kinetics provide reaction rates: r_1 , r_2 , r_3 . Therefore, the reaction rates must first be combined to determine the net production or consumption rate of each species.

Workflow

Reaction Rates



Species Rates



Species Balances



Solve for Concentrations

The most common mistake in multiple-reaction problems is using a reaction rate directly in a mole balance when the net species rate is required instead.

Step 5: Rewrite All Equations in Residual Form

Move everything to one side:

$$f_i(C_A, C_B, C_C, C_D, C_E) = 0$$

You now have a system of nonlinear algebraic equations.

Step 6: Solve the Nonlinear Equation System

After writing the five species balances, you have: **Five Unknowns and Five Equations**

The number of equations equals the number of unknowns, so the problem is mathematically solvable.

Because the reaction rates contain products and powers of concentrations, the resulting equations are nonlinear. Examples include:

CA

CC²*CA

CC*CD

These equations generally cannot be solved one at a time using algebra. Instead, solve them simultaneously using a numerical solver such as Python `scipy.optimize.root` or `scipy.optimize.fsolve`.

Final Answer Checklist

Part 1

- ✓ Calculated all reaction rates.
- ✓ Determined the net species rate.
- ✓ Applied the CSTR mole balance.
- ✓ Calculated the reactor volume.

Part 2

- ✓ Identified five unknown concentrations.
- ✓ Wrote five species balances.
- ✓ Expressed species rates in terms of reaction rates.
- ✓ Solved the nonlinear equation system.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (15) CSTR Mole Balance

Eq. (47) Relative Rates

Key Insight

Reaction Rates



Determine Species Rates



Species Rates



Appear in Mole Balances



Mole Balances



Determine Either Reactor Volume or Exit Concentrations

